JEE OS DATA - extraction review (35 questions with figures)

Copy_of_odea66c2_p046_q009 · Organic Chemistry-Some Basic Principles and Techniques · key **D**

Reaction of hydrogen bromide with propene in the absence of peroxide is a/an

(A) free radical addition

(B) nucleophilic addition

(C) electrophilic substitution

(D) electrophilic addition

extracted text + cropped figure (the JSON)

9. Reaction of hydrogen bromide with propene in the absence of peroxide is a/an

(a) free radical addition

(b) nucleophilic addition

(c) electrophilic substitution

(d) electrophilic addition

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Copy_of_odea66c2_p046_q010 · Organic Chemistry-Some Basic Principles and Techniques · key **A**

One mole of propanone and one mole of formaldehyde are the products of ozonolysis of one mole of an alkene. The alkene may be

(A) 2-methylpropene

(B) 2, 2-dimethyl-1-butene

(C) propene

(D) 2-butene

extracted text + cropped figure (the JSON)

10. One mole of propanone and one mole of formaldehyde are the products of ozonolysis of one mole of an alkene. The alkene may be

(a) 2-methylpropene

(b) 2, 2-dimethyl-1-butene

(c) propene

(d) 2-butene

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Copy_of_odea66c2_p046_q011 · Organic Chemistry-Some Basic Principles and Techniques · key **A**

The compound shown below can be readily prepared by Friedel-Craft's reaction between

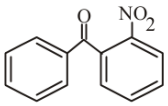
(A) benzene and 2-nitrobenzoyl chloride

(B) benzyl chloride and nitrobenzene

(C) nitrobenzene and benzoyl chloride

(D) benzene and 2-nitrobenzyl chloride

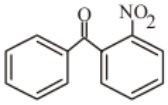
11. The compound shown below



can be readily prepared by Friedel-Craft's reaction between

extracted text + cropped figure (the JSON)

11. The compound shown below



can be readily prepared by Friedel-Craft's reaction between

(a) benzene and 2-nitrobenzoyl chloride

(b) benzyl chloride and nitrobenzene

(c) nitrobenzene and benzoyl chloride

(d) benzene and 2-nitrobenzyl chloride

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Copy_of_odea66c2_p046_q012 · Organic Chemistry-Some Basic Principles and Techniques · key **A**

on mercuration-demercuration produces the major product:

(A) [figure option]

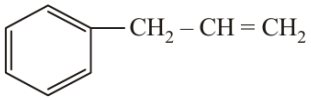
(B) [figure option]

(C) [figure option]

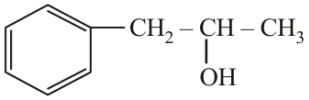
(D) [figure option]

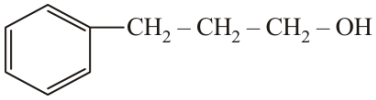
(c) nitrobenzene and benzoyl chloride

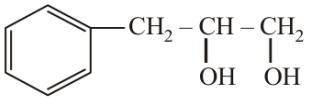
(d) benzene and 2-nitrobenzyl chloride

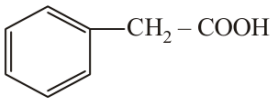
12. 

on mercuration-demercuration produces the major product:

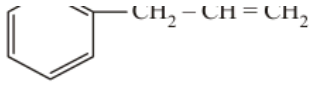
(a) 

(b) 

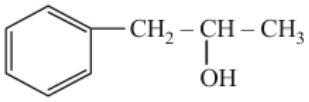
(c) 

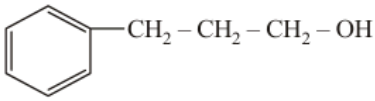
(d) 

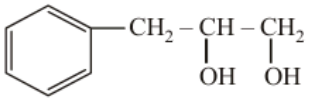
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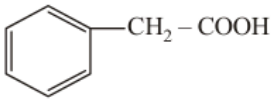
12. 

on mercuration-demercuration produces the major product:

(a) 

(b) 

(c) 

(d) 

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Copy_of_odea66c2_p046_q013 · Organic Chemistry-Some Basic Principles and Techniques · key **A**

In the presence of peroxide, HCl and HI do not give anti-Markownikoff's addition of alkenes because:

(A) One of the steps is endothermic in HCl and HI

(B) Both HCl and HI are strong acids

(C) HCl is oxidizing and the HI is reducing

(D) All the steps are exothermic in HCl and HI

extracted text + cropped figure (the JSON)

13. In the presence of peroxide, HCl and HI do not give anti-Markownikoff's addition of alkenes because:

(a) One of the steps is endothermic in HCl and HI

(b) Both HCl and HI are strong acids

(c) HCl is oxidizing and the HI is reducing

(d) All the steps are exothermic in HCl and HI

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Copy_of_odea66c2_p046_q014 · Organic Chemistry-Some Basic Principles and Techniques · key **B**

The reaction of with HBr gives

(A) [figure option]

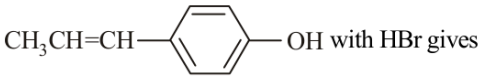
(B) [figure option]

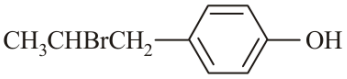
(C) [figure option]

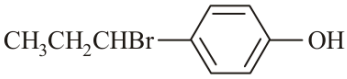
(D) [figure option]

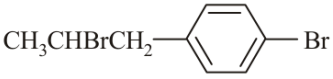
(d) All the steps are exothermic in HCl and HI

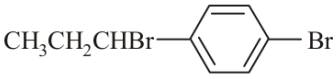
14. The reaction of



(a) 

(b) 

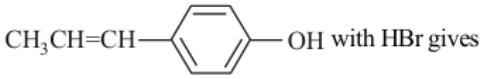
(c) 

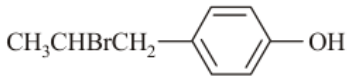
(d) 

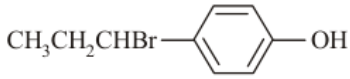
15. Among the following sets, the most stable ionic species are

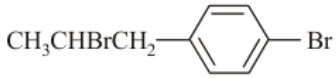
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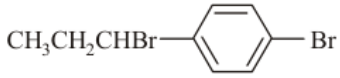
14. The reaction of



(a) 

(b) 

(c) 

(d) 

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Copy_of_odea66c2_p046_q015 · Organic Chemistry-Some Basic Principles and Techniques · key **D**

Among the following sets, the most stable ionic species are

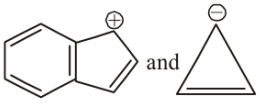
(A) [figure option]

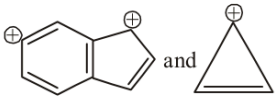
(B) [figure option]

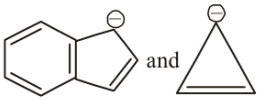
(C) [figure option]

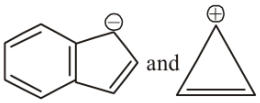
(D) [figure option]

15. Among the following sets, the most stable ionic species are

(a) 

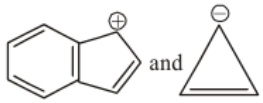
(b) 

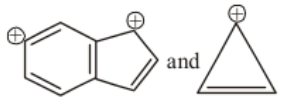
(c) 

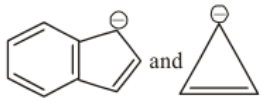
(d) 

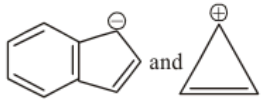
extracted text + cropped figure (the JSON)

15. Among the following sets, the most stable ionic species are

(a) 

(b) 

(c) 

(d) 

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Copy_of_odea66c2_p047_q016 · Hydrocarbons · key **A** FIGURE_CLIPPED

Isopropene, $CH_2=C(CH_3)-CH=CH_2$ is treated with HCl, the products formed will be a mixture of [structures (i)-(iv) drawn: (i) CH3-C(CH3)(Cl)-CH=CH2; (ii) CH2=C(CH3)-CH(Cl)-CH3; (iii) CH3-C(CH3)=CH-CH2Cl; (iv) ClCH2-C(CH3)=CH-CH3]

(A) (i) and (iii)

(B) (ii) and (iii)

(C) (ii) and (iv)

(D) All the four.

extracted text + cropped figure (the JSON)

16. Isopropene, $CH_2=\overset{CH_3}{\overset{|}{C}}-CH=CH_2$ is treated with HCl, the products formed will be a mixture of

(i) $CH_3-\overset{CH_3}{\overset{|}{\underset{Cl}{\underset{|}{C}}}}-CH=CH_2$

(ii) $CH_2=\overset{CH_3}{\overset{|}{\underset{Cl}{\underset{|}{C}}}}-CH=CH_3$

(iii) $CH_3-\overset{CH_3}{\overset{|}{C}}=CH-CH_2Cl$

(iv) $ClCH_2-\overset{CH_3}{\overset{|}{C}}=CH-CH_3$

(a) (i) and (iii) (b) (ii) and (iii)

(c) (ii) and (iv) (d) All the four.

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Copy_of_odea66c2_p047_q017 · Hydrocarbons · key **C**

1, 3-Butadiene is treated with hot $KMnO_4/NaIO_4$, the product(s) formed is(are)

(A) 2HCHO + OHC–CHO

(B) 2HCOOH + HOOC–COOH

(C) $2CO_2$ + HOOC–COOH

(D) $4CO_2$

extracted text + cropped figure (the JSON)

17. 1, 3-Butadiene is treated with hot $KMnO_4/NaIO_4$, the product(s) formed is(are)

(a) $2HCHO + OHC-CHO$

(b) $2HCOOH + HOOC-COOH$

(c) $2CO_2 + HOOC-COOH$

(d) $4CO_2$

Ph Ph Ph Ph MeO

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Copy_of_odea66c2_p047_q018 · Hydrocarbons · key **D** FIGURE_CLIPPED KEY_UNVERIFIED

Order of rate of electrophilic addition reaction with HBr will be

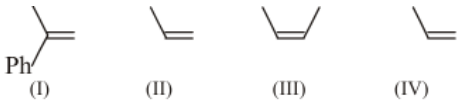
(A) IV > I > III > II

(B) I > II > III > IV

(C) I > III > II > IV

(D) IV > I > II > III

extracted text + cropped figure (the JSON)

18. 

(I) (II) (III) (IV)

Order of rate of electrophilic addition reaction with HBr will be

(a) IV > I > III > II (b) I > II > III > IV

(c) I > III > II > IV (d) IV > I > II > III



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Copy_of_odea66c2_p047_q019 · Hydrocarbons · key **C** FIGURE_CLIPPED KEY_UNVERIFIED

Key C cannot be checked. Printed (a) is the normal Markovnikov product; key C unseen (OCR suggests an alkene-type text, so likely misprint or garbled) - verify against the page.

+ HCl -> X, X is

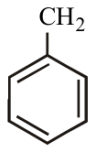
(A) [figure option]

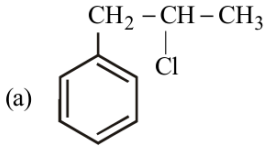
(B) [figure option]

(C) [figure option]

(D) [figure option]

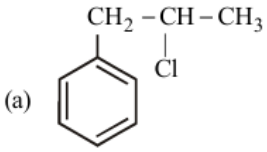
(c) I > III > II > IV (d) IV > I > II > III

19.  + HCl ⟶ X, X is



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19.  + HCl ⟶ X, X is



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Copy_of_odea66c2_p047_q020 · Hydrocarbons · key **A** FIGURE_CLIPPED

The major product of the following reaction is

(A) [figure option]

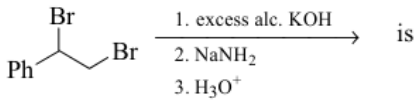
(B) [figure option]

(C) [figure option]

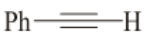
(D) [figure option]

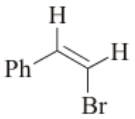
extracted text + cropped figure (the JSON)

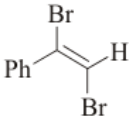
20. The major product of the following reaction

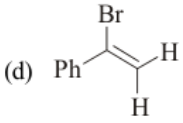


is

(a) 

(b) 

(c) 

(d) 

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Copy_of_odea66c2_p047_q021 · Hydrocarbons · key **B**

The addition of HI in the presence of peroxide catalyst does not follow anti-Markovnikov's rule because

(A) HI is a strong reducing agent.

(B) H-I bond is too strong to be broken homolytically.

(C) I atom combines with H atom to give back HI.

(D) Iodine atom is not reactive enough to add across a double bond.

extracted text + cropped figure (the JSON)

21. The addition of HI in the presence of peroxide catalyst does not follow anti-Markovnikov's rule because

(a) HI is a strong reducing agent.

(b) H-I bond is too strong to be broken homolytically.

(c) I atom combines with H atom to give back HI.

(d) Iodine atom is not reactive enough to add across a double bond.

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Copy_of_odea66c2_p047_q022 · Hydrocarbons · key **D**

Which of the following alcohols cannot be prepared from hydration of an alkene?

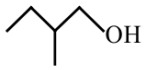
(A) [figure option]

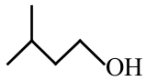
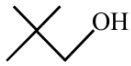
(B) [figure option]

(C) [figure option]

(D) [figure option]

22. Which of the following alcohols cannot be prepared from hydration of an alkene?

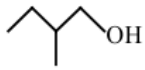
(a)  (b) 

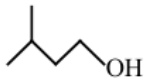
(c)  (d) 

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extracted text + cropped figure (the JSON)

22. Which of the following alcohols cannot be prepared from hydration of an alkene?

(a)  (b) 

(c)  (d) 

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Copy_of_odea66c2_p048_q023 · Hydrocarbons · key **D**

In which of the following reaction, Markovnikov's rule is violated?

(A) $CH_3—O—CH=CH_2$ --(HBr/CCl_4)-->

(B) $CH_3—NH—CH=CH_2$ --(HBr/CCl_4)-->

(C) $CH_3—S—CH=CH_2$ --(HBr/CCl_4)-->

(D) $O_2N—CH=CH_2$ --(HBr/CCl_4)-->

extracted text + cropped figure (the JSON)

23. In which of the following reaction, Markovnikov's rule is violated?

(a) $CH_3—O—CH=CH_2 \xrightarrow[CCl_4]{HBr}$

(b) $CH_3—NH—CH=CH_2 \xrightarrow[CCl_4]{HBr}$

(c) $CH_3—S—CH=CH_2 \xrightarrow[CCl_4]{HBr}$

(d) $O_2N—CH=CH_2 \xrightarrow[CCl_4]{HBr}$

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Copy_of_odea66c2_p048_q024 · Hydrocarbons · key **B**

Hydrogenation of the above compound in the presence of poisoned palladium catalyst gives

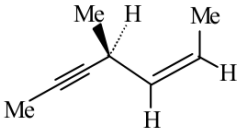
(A) an optically active compound

(B) an optically inactive compound

(C) a racemic mixture

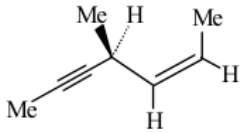
(D) a diastereomeric mixture

(d) $O_2N—CH=CH_2 \xrightarrow[CCl_4]{HBr}$

24. 

Hydrogenation of the above compound in the

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24. 

Hydrogenation of the above compound in the presence of poisoned palladium catalyst gives

(a) an optically active compound

(b) an optically inactive compound

(c) a racemic mixture

(d) a diastereomeric mixture

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Copy_of_odea66c2_p048_q025 · Hydrocarbons · key **C**

The compound X in the following reaction is

$CH_3-C\equiv C-CH_3$ --((i) H_2/Pt, (ii) Br_2)--> X

(A) d-2, 3-Dibromobutane

(B) l-2, 3-Dibromobutane

(C) dl-2, 3-Dibromobutane

(D) meso-2, 3 Dibromobutane.

extracted text + cropped figure (the JSON)

25. The compound X in the following reaction is

$$CH_3C\equiv CCH_3 \xrightarrow[(ii)\ Br_2]{(i)\ H_2\ /\ Pt} X$$

(a) *d*-2, 3-Dibromobutane

(b) *l*-2, 3-Dibromobutane

(c) *dl*-2, 3-Dibromobutane

(d) *meso*-2, 3 Dibromobutane.

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Copy_of_odea66c2_p048_q026 · Hydrocarbons · key **C** FIGURE_CLIPPED

Expected B is the cis alkene. Printed (b) draws the alkene with ring and CH_3 on opposite sides (trans), so the cis alkene must be an unseen option, plausibly C as keyed. Cannot confirm.

--((i) $NaNH_2$, NH_3; (ii) CH_3Br)--> (A) --(H_2, Linder catalyst)--> (B); Product (B) is:

(A) [figure option]

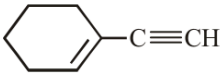
(B) [figure option]

(C) [figure option]

(D) [figure option]

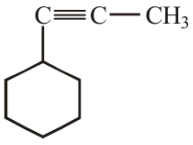
(c) *dl*-2, 3-Dibromobutane

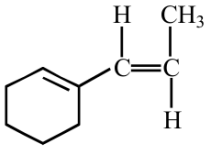
(d) *meso*-2, 3 Dibromobutane.

26. 

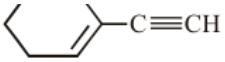
$$\xrightarrow[(ii)\ CH_3Br]{(i)\ NaNH_2,\ NH_3} (A) \xrightarrow[\text{Linder catalyst}]{H_2} (B)\ ;$$

Product (B) is:

(a) 

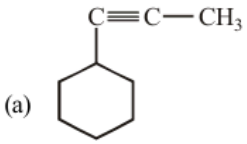
(b) 

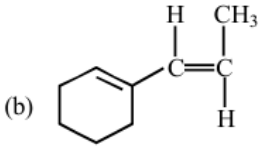
extracted text + cropped figure (the JSON)

26. 

$$\xrightarrow[(ii)\ CH_3Br]{(i)\ NaNH_2,\ NH_3} (A) \xrightarrow[\text{Linder catalyst}]{H_2} (B)\ ;$$

Product (B) is:

(a) 

(b) 

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Copy_of_odea66c2_p048_q027 · Hydrocarbons · key **A**

Among the following compounds, the decreasing order of reactivity towards electrophilic substitution is

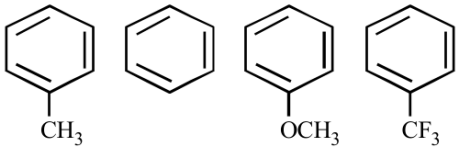
(A) III > I > II > IV

(B) IV > I > II > III

(C) I > II > III > IV

(D) II > I > III > IV

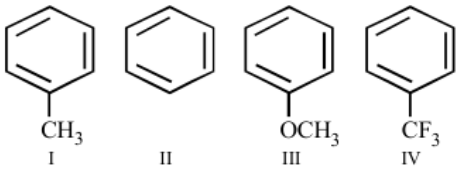
order of reactivity towards electrophilic substitution is



(a) III > I > II > IV (b) IV > I > II > III

extracted text + cropped figure (the JSON)

27. Among the following compounds, the decreasing order of reactivity towards electrophilic substitution is



(a) III > I > II > IV (b) IV > I > II > III

(c) I > II > III > IV (d) II > I > III > IV

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Copy_of_odea66c2_p048_q028 · Hydrocarbons · key **C**

Bromination of ethyl benzene in presence of light gives

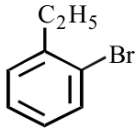
(A) [figure option]

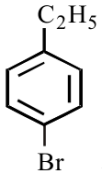
(B) [figure option]

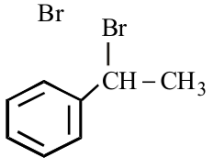
(C) [figure option]

(D) Both (a) and (b)

28. Bromination of ethyl benzene in presence of light gives

(a) 

(b) 

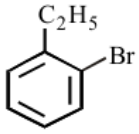
(c) 

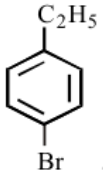
(d) Both (a) and (b)

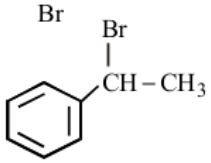
29. Which of the following would not give 2-

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28. Bromination of ethyl benzene in presence of light gives

(a) 

(b) 

(c) 

(d) Both (a) and (b)

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Copy_of_odea66c2_p048_q029 · Hydrocarbons · key **C**

Which of the following would not give 2- phenylbutane as the major product in a Friedel- Crafts alkylation reaction?

(A) 1-butene + HF

(B) 2-butanol + H_2SO_4

(C) Butanoyl chloride + $AlCl_3$ then Zn, HCl

(D) Butyl chloride + $AlCl_3$

extracted text + cropped figure (the JSON)

29. Which of the following would not give 2-phenylbutane as the major product in a Friedel-Crafts alkylation reaction ?

(a) 1-butene + HF

(b) 2-butanol + H_2SO_4

(c) Butanoyl chloride + $AlCl_3$ then Zn, HCl

(d) Butyl chloride + $AlCl_3$

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Copy_of_odea66c2_p049_q030 · Hydrocarbons · key **C**

In the following reaction, P gives two products Q and R, each in 40% yield. If the reaction is carried out with 420 mg of P. the reaction yields 108.8 mg of Q. The amount of R produced in the reaction is closest to

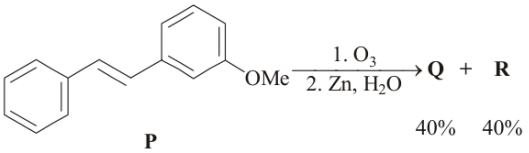
(A) 97.6 mg

(B) 108.8 mg

(C) 84.8 mg

(D) 121.6 mg

30. In the following reaction, **P** gives two products **Q** and **R**, each in 40% yield

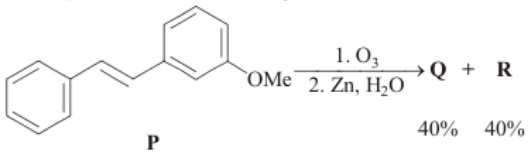


P
(MW = 210)

If the reaction is carried out with 420 mg of **P**.

extracted text + cropped figure (the JSON)

30. In the following reaction, **P** gives two products **Q** and **R**, each in 40% yield



P
(MW = 210)

If the reaction is carried out with 420 mg of **P**. the reaction yields 108.8 mg of **Q**. The amount of **R** produced in the reaction is closest to

(a) 97.6 mg (b) 108.8 mg

(c) 84.8 mg (d) 121.6 mg

Numeric Value Answer

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Copy_of_odea66c2_p049_q031 · Hydrocarbons · key **4**

On conversion into the Grignard reagent followed by treatment with water, how many alkyl bromides would yield isopentane?

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31. On conversion into the Grignard reagent followed by treatment with water, how many alkyl bromides would yield isopentane?

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Copy_of_odea66c2_p049_q032 · Hydrocarbons · key 7 KEY_DISPUTED

solver 3 (conf medium): Wurtz coupling gives butane, pentane, hexane = 3. Book key 7 likely also counts ethane, ethene, propane, propene side products (3+4); keep book key but flag. Haiku right that '

Number of hydrocarbons formed when C_2H_5Br and $CH_3-CH_2-CH_2-Br$ are treated with Na in the presence of dry ether is _____.

extracted text + cropped figure (the JSON)

32. Number of hydrocarbons formed when C_2H_5Br and $CH_3–CH_2–CH_2–Br$ are treated with Na in the presence of dry ether is _____.

								ANSW
1	(c)	**5**	(d)	**9**	(d)	**13**	(a)	**17**
2	(d)	**6**	(a)	**10**	(a)	**14**	(b)	**18**
3	(d)	**7**	(a)	**11**	(a)	**15**	(d)	**19**
4	(b)	**8**	(b)	**12**	(a)	**16**	(a)	**20**

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Copy_of_odea66c2_p049_q033 · Hydrocarbons · key 7

How many number of moles of H2 is used for the complete hydrogenation of the given compound in the presence of a metal catalyst?

the complete hydrogenation of the given compound in the presence of a metal catalyst?



34. Consider the following reaction scheme

extracted text + cropped figure (the JSON)

33. How many number of moles of H_2 is used for the complete hydrogenation of the given compound in the presence of a metal catalyst?

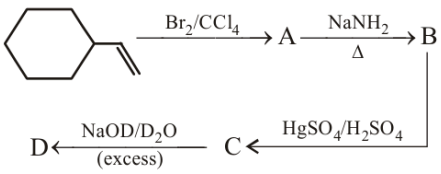


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Copy_of_odea66c2_p049_q034 · Hydrocarbons · key **4**

Consider the following reaction scheme. How many deutrium are present in the compound D?

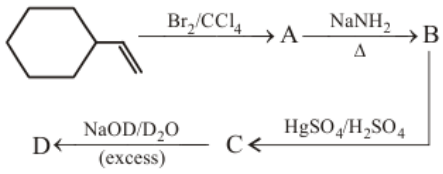
34. Consider the following reaction scheme



How many deutrium are present in the

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34. Consider the following reaction scheme



How many deutrium are present in the compound D?

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Copy_of_odea66c2_p049_q035 · Hydrocarbons · key **3** KEY_DISPUTED

solver 4 (conf high): Triple bond can sit at C1-C2, C5-C6, C6-C7 of heptane chain or in the ethyl group = 4; C2-C3, C3-C4, C4-C5 and the methyl are blocked by CH carbons. Book key 3 looks wrong. Regio

Total no. of alkynes that on catalytic reduction gives 3-ethyl-4-methylheptane _____.

extracted text + cropped figure (the JSON)

35. Total no. of alkynes that on catalytic reduction gives 3-ethyl-4-methylheptane _____.

R KEY								
(c)	**21**	(b)	**25**	(c)	**29**	(c)	**33**	(7)
(d)	**22**	(d)	**26**	(c)	**30**	(c)	**34**	(4)
(c)	**23**	(d)	**27**	(a)	**31**	(4)	**35**	(3)
(a)	**24**	(b)	**28**	(c)	**32**	(7)		

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Copy_of_0dea66c2_p050_q001 · Hydrocarbons · key **A** KEY_DISPUTED

solver D (conf medium): Acrolein/PAN are smog products; chlorine nitrate (ClONO2) is the Antarctic ozone-hole compound. Key A is likely a data-entry error; do not adopt silently.

In Antarctica, ozone depletion is due to the formation of following compound

(A) acrolein

(B) peroxyacetyl nitrate

(C) SO_2 and SO_3

(D) chlorine nitrate

extracted text + cropped figure (the JSON)

1. In Antarctica, ozone depletion is due to the formation of following compound
   (a) acrolein
   (b) peroxyacetyl nitrate
   (c) SO_2 and SO_3
   (d) chlorine nitrate

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Copy_of_0dea66c2_p050_q002 · Hydrocarbons · key **B**

Classical smog occurs in places of

(A) excess SO_2

(B) low temperature

(C) high temperature

(D) excess NH_3

extracted text + cropped figure (the JSON)

2. Classical smog occurs in places of
   (a) excess SO_2 (b) low temperature
   (c) high temperature (d) excess NH_3

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Copy_of_0dea66c2_p050_q003 · Hydrocarbons · key **B**

Which gas is responsible for ' Bhopal Gas Tragedy' in 1984?

(A) CO

(B) Methyl isocyanate

(C) SO_2 and NO_2

(D) Ethyl isocyanate

extracted text + cropped figure (the JSON)

3. Which gas is responsible for ' Bhopal Gas Tragedy' in 1984?
   (a) CO
   (b) Methyl isocyanate
   (c) SO_2 and NO_2
   (d) Ethyl isocyanate

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Copy_of_0dea66c2_p050_q004 · Hydrocarbons · key **B**

CFC which is a main reason behind air pollution, is produced by

(A) sewage pollutant

(B) aerosols

(C) industrial remains

(D) all the these

extracted text + cropped figure (the JSON)

4. CFC which is a main reason behind air pollution, is produced by
   (a) sewage pollutant
   (b) aerosols
   (c) industrial remains
   (d) all the these

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Copy_of_0dea66c2_p050_q005 · Hydrocarbons · key **C**

Which is a dangerous radiological pollutant?

(A) C^{14}

(B) S^{35}

(C) Sr^{90}

(D) P^{32}

extracted text + cropped figure (the JSON)

5. Which is a dangerous radiological pollutant?
   (a) C^{14} (b) S^{35}
   (c) Sr^{90} (d) P^{32}

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Copy_of_odea66c2_p050_q006 · Hydrocarbons · key **B**

Which of the following is the coldest region of atmosphere?

(A) Thermosphere

(B) Mesosphere

(C) Troposphere

(D) Stratosphere

extracted text + cropped figure (the JSON)

5. Which of the following is the coldest region of atmosphere?
   (a) Thermosphere (b) Mesosphere
   (c) Troposphere (d) Stratosphere

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Copy_of_odea66c2_p050_q007 · Hydrocarbons · key **A**

Green house gases –

(A) allow shorter wavelength to enter earth's atmosphere while doesn't allow longer wavelength to leave the earth's atmosphere.

(B) allow longer wavelength to enter earth atmosphere while doesn't allow shorter wavelength to leave the surface

(C) don't have wavelength specific behaviour.

(D) show wavelength specific behaviour near the earth while far from earth these have wavelength independent behaviour.

extracted text + cropped figure (the JSON)

7. Green house gases –
   (a) allow shorter wavelength to enter earth's atmosphere while doesn't allow longer wavelength to leave the earth's atmosphere.
   (b) allow longer wavelength to enter earth atmosphere while doesn't allow shorter wavelength to leave the surface
   (c) don't have wavelength specific behaviour.
   (d) show wavelength specific behaviour near the earth while far from earth these have wavelength independent behaviour.

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Copy_of_odea66c2_p050_q008 · Hydrocarbons · key **D**

Formation of London smog takes place in

(A) winter during day time

(B) summer during day time

(C) summer during morning time

(D) winter during morning time

extracted text + cropped figure (the JSON)

8. Formation of London smog takes place in
   (a) winter during day time
   (b) summer during day time
   (c) summer during morning time
   (d) winter during morning time

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